

MATERIALS, PROCESSES AND SURFACE PROTECTIONS

Resumen/Summary

This Note includes the selection of the materials, processes and surface protection that students can use in design of the Elevator for the Master Thesis and Exercises.

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1. MATERIALS

This chapter includes the selection of metallic and no-metallic materials to be used in the Elevator.

When ever possible, it will be minimized the use hazardous materials.

1.1 METALLIC MATERIALS

1.1.1 Aluminium alloys

Standard designation				2024 T3						2024 T42			
Usual designation				Aluminium alloy alclad									
Density		kg/dm ³		2.77									
Use information				For skin and formed parts									
Form				Sheets									
Delivery condition				T3						T3 or O			
Use condition				T3						T42			
Limit dimensions		mm		0,6 ≤ t ≤ 1,6		1,6 < t ≤ 3,2		3,2 < t ≤ 6,32		0,6 < t ≤ 1,6		1,6 < t ≤ 6,32	
Direction of test piece				L	LT	L	LT	L	LT	L	LT	L	LT
Characteristics at 20°C	Tensile strength	R _m	MPa	414	407	428	421	434	428	393	393	414	414
	Yield strength	R _{p0,2}	MPa	303	269	310	276	310	276	235	235	248	248
	Elongation	A	%	-	15	-	15	-	15	-	15	-	15
	Modulus	E	MPa	72418									
Procurement specification				AMS-QQ-A-250-5									

Aluminium alloys (cont')

Standard designation				7050T7451													
Usual designation				Aluminium Alloy													
Density		kg/dm ³		2,82													
Use information				N/A													
Form				Plates													
Delivery condition				T7451													
Use condition				T7451													
Limit dimensions		mm		t ≤ 51		51 < t ≤ 76			76 < t ≤ 102			102 < t ≤ 127			127 < t ≤ 152		
Direction of test piece				L	LT	L	LT	ST	L	LT	ST	L	LT	ST	L	LT	ST
Characteristics at 20° C	Tensile strength	R _m	MPa	510	510	503	503	469	496	496	469	490	490	462	483	483	462
	Yield strength	R _{p0,2}	MPa	441	441	434	434	407	427	427	400	421	421	393	414	414	393
	Elongation	A	%	9	8	8	7	2	8	5	2	8	5	2	7	4	2
	Modulus	E	MPa	71039													
Procurement specification				AMS 4050 + K _{IC} Requirements													

1.1.2 Stainless steels

Standard designation				PH 13 – 8Mo –H1050	
Usual designation				PH Stainless Steel	
Density		kg/dm ³	7.72		
Use information				N/A	
Form				Rectangular Round Rod and Bars	
Delivery condition				SLHT (Solution Heat Treated)	
Use condition				H1050	
Limit dimensions		mm	Thickness or Diameter ≤ 305		
Direction of test piece				L and LT	
Characteristics at 20° C	Tensile strength	R _m	MPa	1207	
	Yield strength	R _{p0,2}	MPa	1138	
	Elongation	A	%	11	
	Modulus	E	MPa	195185	
Procurement specification				AMS 5629	

1.1.3 Titanium alloys

Standard designation				Ti 6 Al 4V – ANLD					
Usual designation				Titanium alloy					
Density		kg/dm ³		4,43					
Use information				N/A					
Form				Sheets		Plates		Forgings	
Delivery condition				Annealed					
Use condition				Annealed					
Limit dimensions Thickness		mm		t < 50		50 ≤ t ≤ 100		100 < t ≤ 150	
Direction of test piece				L and LT		L, LT and ST		L and LT ST	
Characteristics at 20° C	Tensile strength	R _m	MPa	931		897		897 897	
	Yield strength	R _{p0,2}	MPa	862		828		828 828	
	Elongation	A	%	10		10		10 8	
	Modulus	E	MPa	110352					
Procurement specification				AMS 4911		AMS 4911		AMS 4928	

1.1.4 Copper Alloys

Standard designation				C63000ANLD-ROD (add diameter to uase)			
Usual designation				Copper Alloys Rods (Niquel Aluminum Bronze)			
Density		kg/dm ³		8,9			
Use information				Used for Bushings			
Form				Rods			
Delivery condition				Annealed			
Use condition				Annealed			
Limit dimensions Thickness		mm		Diameter ≤ 110			
Direction of test piece				L and LT			
Characteristics at 20° C	Tensile strength		R _m	MPa			
	Yield strength		R _{p0,2}	MPa			
	Elongation		A	%			
	Modulus		E	MPa			
Procurement specification				AMS 4640			
Dassault Aviation agreement							

1.2 NON-METALLIC MATERIALS

The following non metallic materials have been selected a a basis for the Falcon SMS Horizontal and Vertical Stabilizer design.

1.2.1 ANTICORROSIVE PROTECTIONS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE OF MATERIAL	CHARACTERISTIC/REMARKS
Z-11.906	AIMS04-05-005 Type I	Corrosion inhibiting compound for removable fasteners protection against corrosion. Service temperature <90°	Polysulphide based ant-corrosive compound, used to seal and provide corrosion inhibition on assenblies which need to be easily removed during their servive life. Service Temperature: -55 to 95°C Density: 1,35 g/cm ³
Z-11.909	AIMS04-05-005 Type II	Corrosion inhibiting compound for removable fasteners protection against corrosion. Service temperature > 90°	Polysulphide based ant-corrosive compound, used to seal and provide corrosion inhibition on assenblies which need to be easily removed during their servive life. Service Temperature: -55 to 200°C Density: 1,0 g/cm ³

1.2.2 PAINTS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE OF MATERIAL	CHARACTERISTIC/REMARKS
Z-12.105	AIMS 04-04-012	Polyurethane primer	Two-parts polyurethane primer. Used as single coat for interior protection or as base for polyurethane finishes.
Z-12.113	AIMS 04-04-001	Epoxy primer	Two-parts epoxy primer
Z-12.129	AIMS 04-04-002	Epoxy-urethane primer Chromate-free	Two-parts epoxy/urethane primer, without chromates, phosphoric ester based hydraulic fluids (Skydrol) resistant. Used on composite materials, as single coat for protection of interiors or as base for polyurethane finishes.
Z-12.210	Commercial	Epoxy Filler	Two parts epoxy filler, for reinforced plastic parts or metallic surfaces. Pot life: 30 minutes at 25°C. Curing: 24 hours at r.t. or 1 hour at 70°C.
Z-12.211	AIMS 04-04-011	Polyurethane Pore-filler	Two-parts polyurethane pore-filler, for application on composite materials (glass, carbon or aramid fiber fabrics, with epoxy resins) prior to application of polyurethane paint systems. Density (cured): 1.45 g/cm ³
Z-12.380	AIMS 04-04-003	Polyurethane coating (Interiors)	Two-parts polyurethane coating, compatible with epoxy or polyurethane primers, resistant to lubricants up to 120°C, to phosphoric ester based hydraulic fluids (skydrol) and to synthetic greases and oils.
Z-12.402	AIMS 04-04-009	Blue coating for grounding identification	Two-parts direct adhesion, epoxy paint, for identification purposes in electrical bonding. Colour: blue
Z-12.412	AIMS 04-04-023	Polyurethane varnish	Two-parts polyurethane varnish, clear, colourless, resistant to phosphoric ester base hydraulic fluids (skydrol), u.v. radiations and to abrasion.
Z-12.524	AIMS 04-04-010	Sealant bead protection paint	Two-parts direct adhesion, epoxy paint, for protection of rivet lines and sealant fillets against hydraulic fluid projections. mat. 1 is also used for identification purposes in electrical bonding. Color: grey (BAC 707)
Z-12.607	AIMS 04-04-012	Thinner	General thinners for polyurethane organic coatings.

1.2.3 PLASTIC MATERIALS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE MATERIAL	OF CHARACTERISTIC/REMARKS
Z-14.301	L-P-410A Nylon 6/6	Polyamide 6/6 rigid	Polyamide 6/6 (polyhexamethylene-adipamide) in sheets, tubings or bars. Specific gravity: 1.14 ± 0.01 Deflection temperature under load: With 0.45 MPa: 182°C minimum. With 1.80 MPa: 66°C minimum. Applicable semi-finished part standards: LN-9392 (Sheets, thicknesses 1 to 6 mm) LN-9388 (Rods). LN-9393 (Tubings, thick wall). LN-29980 (Tubings, thin wall). LN-29981 (Fat section rods)
Z-14.510	L-P-1040B Grade B Class 1	Polyvinyl fluoride film	Polyvinyl fluoride film (pvf), bondable on both sides. Colour: Granite grey, medium gloss (BAC 705). Thickness: 50 µm. Areal weight: 77 g/m².

1.2.4 ADHESIVES

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE OF MATERIAL	CHARACTERISTIC/REMARKS
Z-15.201	I+D-N-200	Epoxy adhesive (EC-2216)	Two-parts epoxy adhesive, for relatively flexible joints in non-structural bondings. Service temperatures: -55 to 80°C. Curing: 3-7 days at R.T or 2 hours at 65-70°C.
Z-15.213	I+D-N-200	Epoxy adhesive (Araldit 420)	Two-parts epoxy adhesive, thixotropic, for general use and for attachment of inserts. Mixing ratio: 100 parts by weight of base and 40 of hardener. Pot life: 1 hour at 25°C. Curing: 7 days at R.T. or 1 hour at 120°C. Density (Cured): 1.15 ± 0.05 g/cm ³
Z-15.237	I+D-N-200	Epoxy adhesive (EA-934)	Two-parts epoxy adhesive for structural metal/metal bondings, liquid shims (as per I+D-P-61) aerodynamic fillers, etc Low peel strength. Service temperature: -55 to 150°C. Curing: 5-7 days at R.T or 1 hour at 93°C.
Z-15.415	AIMS 10-02-001 Grade B	Foam adhesive	Epoxy foam adhesive, for cores filling and potting, and for core/core and core/edge joints. Areal weight (g/m ²): 1000-1600. Nominal thickness: 1.27 mm (0.050"). Expansion ratio: 1.5 to 3.5. Curing: 90 minutes at 115°C or 60 minutes at 170°C.
Z-15.425	AIMS 10-01-006 Grade A	Film adhesive EA-9695.03NW FM-300M0.3	Epoxy film adhesive, with polyester mat carrier, for general use in composite bondings, metal/metal bondings or a combination of them. Service temperature: -55 to 135°C. Areal weight: 170 ± 25 g/m ² Nominal thickness: 0.13 mm. Curing temperature: 180 ± 5°C. Associated primer: Z-15.915.
Z-15.467	AIMS 10-01-005 Grade A	Film adhesive EA-9695.03NW	Epoxy film adhesive, with polyester mat carrier, for general use in composite bondings, metal/metal bondings or a combination of them. Specially recommended as surfacing adhesive in cocuring processes for sandwich structures. Service temperature: -55 to 120°C. Areal weight: 150 ± 15 g/m ² . Thickness: 0.150 mm (Nominal). Curing temperature: 175 ± 5°C. Associated primer: Z-15.915.
Z-15.468	AIMS 10-01-005 Grade B	Film adhesive EA-9695.05K	Epoxy film adhesive, with polyester knit carrier, for general use in composite bondings. Specially recommended for manufacturing of sandwich structures by cocuring or cobonding processes. Service temperature: -55 to 120°C. Areal weight: 245 ± 20 g/m ² . Thickness: 0.250 mm (Nominal). Curing temperature: 175 ± 5°C. Associated primer: Z-15.915

1.2.5 SEALANTS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE OF MATERIAL	CHARACTERISTIC/REMARKS
Z-16.171	AIMS04-05-014 Class C	Faying surface sealant Wet Install fasteners Bushings Installations Bearings Installations	Polysulfide sealant for faying surface sealing in areas requiring long application times. Application: Brush, spatula or extrusion gun. Density (cured): 1.4 g/cm ³ max.
Z-16.176	AIMS04-05-014 Class A	Fasteners heads and collars protection	Chromate free Polysulfide sealant with high aviation fuel and lubricant oil resistance. Overcoat use. Service temperature: -55 to 135°C. Density (cured): 1.15 g/cm ³ max.
Z-16.177	AIMS04-05-014 Class B	Fillet seal. Sealant beads with demoulding agent on removable side and easy removable/installation path.	Chromate free Polysulfide sealant, with high aviation fuel and lubricant oils resistance. Fillet use. Service temperature: -55 to 135°C Hardness Shore A: 50 minimum Density (cured): 1.15 g/cm ³ max.
Z-16.160	AIMS04-05-006	Faying surfaces sealant. Low adhesion. Removable joints	Polysulfide (permapol P-5) based sealant, magnesium bixide cured, low adhesion, for sealing dismountable elements in general. Service temperature: -55 to 120°C. Application: Brush Density (cured): 1.6 g/cm ³ max.
Z-16.161	AIMS04-05-006	Fillet seal. Low adhesion. Removable joints	Polysulfide (permapol P-5) based sealant, magnesium bixide cured, low adhesion, for sealing dismountable elements in general. Service temperature: -55 to 120°C. Application: Spatula or extrusion gun Density (cured): 1.6 g/cm ³ max.

1.2.6 DRY FABRICS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE MATERIAL OF	CHARACTERISTIC/REMARKS
Z-17.214	Comercial 07791 M 1270 TF970 E01 2F	Glass fiber fabric for RTM	Glass fiber fabric (Z17.208), with epoxy binder on one face, used as surface protection by the Resin Transfer Moulding (RTM) process with the resin Z18173. • Texture: 8 harness satin. • Areal weight: 295 g/m² • Thickness: 0.23 mm
Z-17.237	AIMS 05-04-009	Carbon fiber fabric	Carbon fiber fabric HTA-6K with epoxy binder, for use in the resin transfer moulding (RTM) process, with the resin RTM-6 (Z-18.173). Texture: 5 harness satin. Fabric areal weight: 370 g/m ² . Areal weight (Incl. Epoxy binder): 390 g/m ² . Note: The laminate manufactured with this fabric and the Z-18.173 resin will have 57% fiber volume and 0.370 mm thickness per ply.
Z-17.251	AMS-C-9084T.III C2	Glass fiber fabric for RTM	Type 120 glass fiber fabric (Z17.201), finish Z6040 and with epoxy binder on one face, used as surface protection by the Resin Transfer Moulding (RTM) process with the resin Z18173. • Texture: 4 harness satin. • Areal weight: 104 ± 6 g/m² • Thickness: 0,09 – 0,14 mm

1.2.7 HONEYCOMB CORES

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE MATERIAL	OF CHARACTERISTIC/REMARKS
Z-17.608	ABS 5035 A5	Aramid honeycomb core 3.0-3/16	Aramid honeycomb core, impregnated in phenolic resin, hexagonal cell. Cell size: 3/16" (4.8 mm). Density: 3.0 lb/ft³ (48 kg/m³).

1.2.8 RESINS AND FILLERS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE MATERIAL	OF	CHARACTERISTIC/REMARKS
Z-18.144	I+D-N-200	Epoxy resin		Two-parts epoxy resin, thixotropic, room temperature curing, for filling, potting and stabilizing of honeycomb cores. Service temperature: -55 to 80°C. Pot life (200 g): 90 minutes. Curing: 7 days at R.T. or 5 hours at 50°C. Density (cured): 0.45 g/cm³ maximum.
Z-18.173	AIMS 05-04-100	Epoxy resin RTM 6		One-part epoxy resin, developed to use in advanced resin transfer moulding (RTM) processes, with the carbon fabric Z-17.237 or Z-17.244. Working life: 15 days at 23°C. Curing: 90 minutes minimum at 180 ± 5°C.

1.2.9 PREPREGS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE MATERIAL OF	CHARACTERISTIC/REMARKS
Z-19.102	I+D-N-200	Glass/Epoxy prepreg	Glass fiber fabric Z-17.208 (Type 181) impregnated in epoxy resin (MIL-R-9300 Types I & II) Areal weight: $460 \pm 20 \text{ g/m}^2$ Resin content: $36 \pm 2\%$ Curing temperature: $177 \pm 5^\circ\text{C}$. Thickness per ply: $0.230 \pm 0.030 \text{ mm}$
Z-19.113		Quartz/Epoxy prepreg	Quartz glass fiber (quartz silica 300/2-2) fabric, preimpregnated in epoxy resin M14. Service Temperature: -55 to 130°C Areal weight : $505 \pm 25 \text{ g/m}^2$ Resin content: $43 \pm 3\%$ Curing temperature: $185 \pm 5^\circ\text{C}$ Thickness per ply: $0.290 \pm 0.015 \text{ mm}$
Z-19.304	I+D-P-250A Group II	Glass/Epoxy prepreg	Glass fiber fabric Z-17.201 (type 120) impregnated in epoxy resin. Areal weight: $177 \pm 25 \text{ g/m}^2$. Resin content: $42 \pm 3\%$. Curing temperature: $180 \pm 5^\circ\text{C}$. Thickness per ply: $0.100 \pm 0.005 \text{ mm}$.
Z-19.760	AIMS 05-01-003 Group V	Carbon/Epoxy prepreg (Fabric) (AGP 193 PW/8552)	Carbon fiber fabric AGP193 PW impregnated in epoxy resin. Areal weight: $325 \pm 25 \text{ g/m}^2$. Resin content: $40 \pm 2\%$. Curing temperature: $180 \pm 5^\circ\text{C}$. Thickness per ply: $0.205 \pm 0.010 \text{ mm}$.
Z-19.776	I+D-P-230B Group IV	Carbon/Epoxy prepreg (Fabric) W3T-282-42-F593	Carbon fiber fabric Z-17.225 preimpregnated in epoxy resin. Areal weight: $345 \pm 25 \text{ g/m}^2$. Resin content: $44 \pm 2\%$. Curing temperature: $180 \pm 5^\circ\text{C}$. Thickness per ply: $0.237 \pm 0.017 \text{ mm}$.
Z-19.780	AIMS 05-01-001	Carbon/Epoxy prepreg (Tape) AS4/8552 RC34 AW194 Standard Modulus	Carbon fiber tape impregnated in epoxy resin. Areal weight: $297 \pm 15 \text{ g/m}^2$. Resin content: $34 \pm 2\%$. Curing temperature: $185 \pm 5^\circ\text{C}$. Thickness per ply: $0.184 \pm 0.010 \text{ mm}$.
Z-19.778	AIMS 05-01-002 Grade A	Carbon/Epoxy prepreg (Tape) M21/34%/194/T800S-24K-WIDTH	Carbon fiber tape intermediate modulus (IMS24 or T800S) impregnated in epoxy resin. Areal weight: $194 \pm 8 \text{ g/m}^2$. Resin content: $34 \pm 2\%$. Curing temperature: $185 \pm 5^\circ\text{C}$. Thickness per ply: $0.184 \pm 0.010 \text{ mm}$.

3.12 MISCELLANEOUS

DRAWING DESIGNATION	PROCUREMENT SPECIFICATION	TYPE MATERIAL	OF CHARACTERISTIC/REMARKS
Z-22.949	AIMS 05-12-001	Bronze mesh prepreg	Bronze mesh prepimpregnated in epoxy resin, for lightning strike protection in reinforced plastic parts. Service temperature: -55° to 120°C Bronze mesh fabric weight: 80 ± 5 g/m ² Resin content: 27 ± 3% Curing temperature: 185 ± 5°C

2. MANUFACTURING PROCESS AND NDT SPECIFICATION

This chapter includes the preliminary selection of manufacturing process and NDT specifications to be used in the Falcon SMS Horizontal and Vertical Stabilizer.

When ever possible, it will be minimized the use hazardous products, preparations or substances during the manufacturing process.

PROCESS	EADS-CASA Standard	REMARKS
Shot peening	I+D-P-06	
Degreasing with stabilized organic solvents in vapour phase	I+D-P-09	
Passivation of corrosion-resistant steel	I+D-P-38	
Manufacturing of resin impregnated glass fiber reinforced plastic.	I+D-P-52	
Dry lubrication (molybdenum disulfide base)	I+D-P-58	
Paint (General specification)	I+D-P-60	
Liquid shims	I+D-P-61	
Paint systems for aeronautical detail parts and assemblies	I+D-P-64	
Non-structural bondings (General specification)	I+D-P-71	
Cleaning and pickling of aluminium and its alloys	I+D-P-72	
Abrasive cleaning	I+D-P-74	
Cleaning and pickling of titanium and titanium alloys	I+D-P-90	
Adherence tape test on organic coating	I+D-E-103	
Aluminium alloy and niobium titanium alloy solid shank rivets	I+D-L-113	
Electrolytic cadmium plating	I+D-P-115	
Steel and titanium Hi-Lok fasteners	I+D-P-117	
Magnetic particle inspection process	I+D-E-136	
Penetrant inspection process	I+D-E-137	

PROCESS	EADS-CASA Standard	REMARKS
Measurement of the electric conductivity through induced current.	I+D-E-140	
Application of the sealant	I+D-P-146	
Manual cleaning with solvents	I+D-P-147	
Chemical milling of aluminium and aluminium alloys	I+D-P-152	
Storage and handling of elastomeric products	I+D-N-171	
Fusion welding of metals	I+D-P-181	
Sealant protection against projections of hydraulic fluids	I+D-P-186	
Chemical conversion films on aluminium and its alloys.	I+D-P-202	
Forming, straightening and fitting of metallic parts	I+D-P-206	
Cold working of holes	I+D-P-209	
Forming and straightening of aluminium and its alloys	I+D-P-212	
Anodizing of aluminium and its alloys	I+D-P-213	To agree.
Heat treatment of the aluminium alloys	I+D-P-220	
Heat treatment of titanium and its alloys	I+D-P-222	
Heat treatment of stainless and refractory steels	I+D-P-226	
Riveting of structures of carbon fiber. Composite materials	I+D-P-231	
Manufacturing of carbon composite material structures (laminated and sandwich)	I+D-P-233	
Non-destructive inspection of carbon fiber composite parts	I+D-E-275	
Ultrasonic testing	I+D-E-276	
Ultrasonic pulse-echo inspection of carbon fibre composite parts	I+D-E-280	
Through-transmission ultrasonic inspection of carbon fiber composite parts	I+D-E-282	
Tap testing on bonded elements and/or composite structures.	I+D-E-294	
Reticulation tests in epoxy and polyurethane primers	I+D-P-299	
Radiographic inspection of composite material structures	I+D-E-303	
Swaging of permaswage sleeves on tubes intended for fluid systems.	I+D-P-312	
Erosion resistant paint system	I+D-P-313	

PROCESS	EADS-CASA Standard	REMARKS
Composite laminates bearing test on fastener joints	I+D-E-325	
Application of antistatic paints on reinforced plastic surface.	I+D-P-337	
Bonding and grounding installation	I+D-P-355	
Blind rivets low and medium strength	I+D-P-356	
Painting of carbon fiber composite material parts.	I+D-P-358	To agree
Bush installation in interference assemblies	I+D-P-366	
Wire locking of parts	I+D-P-373	
Locking of screwed connection	I+D-P-376	
Application of corrosion preventive temporary protective compounds	AIPS 02-04-001	
The manufacture of composite components by the resin transfer moulding process	AIPS 03-02-009	

3. SURFACE PROTECTIONS

The purpose of this chapter is to define the preliminary selection of the surface. When ever possible, it will be minimized the use hazardous products and substances during the applications of surfaces protections.

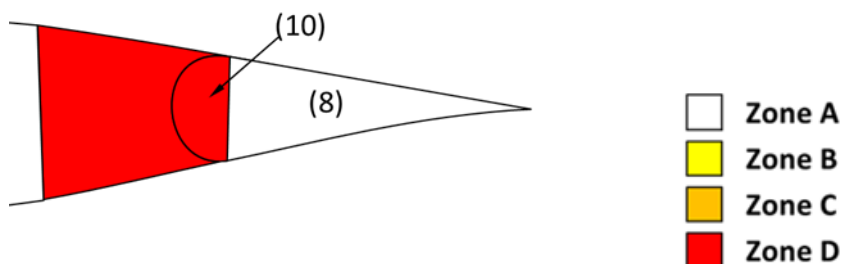
3.1 ENVIRONMENTAL EXPOSURE ZONES

The protection schemes depend, besides the material, on the different environmental exposure zones (and on their corrosion susceptibility. From the point of view of corrosion susceptibility, the following areas of HTP Trailing Edge and Elevator stand out:

Area A: Zone in contact with air, in general dry and with easy access.

Area C: Areas exposed to humidity, water accumulation, areas of difficult access and occasionally contaminated by hydraulic fluids and other aircraft fluids (e.g. toilet, control surfaces, etc.)

Area D: Especially critical areas within area C where water, moisture and other liquids are likely to collect and areas with difficult access.



Elevator zone inside the box (8) is zone A if FS inspection holes are covered (at least with adhesive film)

3.2 TABLES OF PROTECTIVE TREATMENTS

3.2.1 Aluminium alloys

Aluminum parts:

- Zone A:XFU = XXQ + XDV = TSA (tartaric sulfuric anodising)+ P (epoxyprimer)
- Zone D:XFV = XXQ + XDV + XEZ = TSA (tartaric sulfuric anodising)+ P (epoxyprimer) + TC (polyurethaneenamel)

Primer:Z12.113.

Top Coat: Z12.380 y Z12.391*

* Para superficies de aluminio protegidas con TSA se puede utilizar en lugar de Z12.380 y Z12.391, las pinturas de base agua Z12.362y Z12.363, teniendo en cuenta que las imprimaciones aplicables a estas pinturas son Z12.142 y Z12.147 respectivamente.

- Al surface:CCC (chemical conversion coating) =XAA on holes in Al for b

3.2.2 Stainless steels

JOINTS

Fixed bushings into Al parts:

CRES:(Zn.Ni/Cadmium plated) -> AIPS 02-04-006 Electrolytic Zinc-Nickel plating Type II (TNA-007-10003 codeRA004)

Al surface:CCC (chemical conversion coating) =XAA

CRES Bushings (no contact with AL): XCD -> passivated.

Bronze Bushings: XXX -> no treatment

Material Origin and use	CODE (1)	AREA	PROTECTIVE TREATMENT		REMARKS
			Basic	Before Assembly	
Heat treated machined austenitic steel parts, in contact with others, titanium and other types of steels	XCC	E	Passivation as per I+D-P-38 Type I	None	(6)
Welded, heat treated, machined martensitic ferrous steels parts, in contact with others, titanium and other types of steels	XCD	E	Passivation as per I+D-P-38 Type II	None	(6)
Parts or bushings in contact with aluminium and aluminium alloys	XCP	E	Cadmium plated as per I+D-P-115	None	(7) to agree
Parts in contact with CFRP	XVF or XXG	E	Passivation as per I+D-P-38 Types I or II	PAC + F	(4) (5) (6)

3.2.3 Titanium alloys

Tiparts (ACTR fittings)in contact with aluminum or steel:

- Zone D:XTN = XDR + XDV + XEZ = Anodized + primer (epoxyprimer) + TC (polyurethaneenamel)

(requirement in NT due to aluminum contact in a zone with moisture)

Material Origin and use	CODE (1)	AREA	PROTECTIVE TREATMENT		REMARKS
			Basic	Before Assembly	
Parts in contact with aluminium or steel with service temperature up to 120°C	XWX	E	OAT	PAC	(3) (4)

3.2.4 Composite materials

Material Origin and use	CODE (1)	AREA	PROTECTIVE TREATMENT		REMARKS
			Basic	Before Assembly	
Glass fibre or tedlar coated carbon fibre parts in contact with aluminium	XXX	E	None		
Composite components parts inside the aircraft with surface protection not called out in the drawing	XXX	E	None		
Composite components parts which form part of the outside aircraft surface without conductive requirements or with conductive coat on exterior surface	XNC	E	Primer		(10)
Composite parts with conductivity requirements, excluding CFC. Aircraft exterior, dynamic pressure zone. Internal surface	XXX	E	None		
CFC parts without insulation layers in contact with aluminium Internal surfaces	XVA	E-P	Primer	F	(10)

3.2.5 Assemblies

Material Origin and use	CODE (1)	AREA	PROTECTIVE TREATMENT		REMARKS
			Basic	Before Assembly	
Grounding zone	XWS	E-P	None	Cover with sealing compound Z-16.104	
Grounding zone	XMS	E-P	None	Cover with paint Z-12.402	
Rivets rows of visible fasteners surfaces, without rivet heads on interior surface to be covered only with sealant	XXE	E	PAC		(4)

3.2.6 Remarks on the tables of protective treatments

Meaning of the cross-reference numbers appearing in the "Remarks" column of the Tables of protective treatments.

- (1) Internal Airbus code.
- (2) TSA = Tartaric Sulfuric Anodizing unsealed (to a)
- (3) OAT = Sulfuric anodizing of titanium alloys as per I+D-P-48 Class B.
- (4) PAC = Polyurethane anticorrosion primer
- (5) F = Polyurethane finish .
- (6) Do not perform passivation after assembly or after tack welding or seam welding (risk of acid retention) or on pipes containing components likely to retain acid (ball-joints.....).
- (7) Cadmium plating must mandatorily be followed by de-embrittlement, according to I+D-P-115. (to agree)
- (8) Contact between fuel and cadmium is strictly prohibited.
- (9) Abrasive cleaning as per I+D-P-74 before painting.
- (10) The solid film lubricant may extended beyond the limits of the useful area (to be specified on drawing).
- (11) Z-12.129 is a chromate-free primer.